Home / Grade 10 / Geometry / Quarter 4

Annual control work, and the Grade 11 preview

The whole year in one paper, then a look at the road that leads out of it.

LESSON 67–68 | 2 × 40 MIN | GEOMETRY 10, YILLIK NAZORAT ISHI | ANNUAL REVIEW

LESSON CLOCK

3 | 45' | 12' | 12' | 8'

Instructions	3 min
The paper	45 min
Answers	12 min
Diagnosis	12 min
The Grade 11 preview	8 min

LEARNING OBJECTIVES

- Apply the axiomatic, parallel, perpendicular and coordinate methods of the whole year.
- Choose the shortest route to each answer under time pressure.
- Classify each lost mark and rewrite the solution.
- See how Grade 11 continues each strand of Grade 10.

TEXTBOOK

National	pp. 277–280
Cambridge	Pure Mathematics 1, pp. 85–88
Workbook	Annual paper G

01 Explanation

The paper — 40 marks, 45 minutes

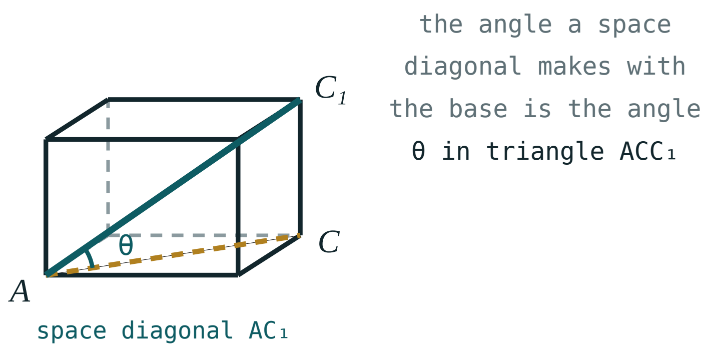
Q	TASK	MARKS	FROM
1	State the three positions of two lines in space, with a sketch of each	*4*	Q I
2	Prove that a line parallel to a line of a plane is parallel to the plane, or lies in it	*6*	Q II
3	A cube of edge 8: the angle between the space diagonal and the base, and between the plane ABC_1 and the base	*7*	Q III
4	Construct and name the section of a cube through three given vertices; find its area	*7*	Q III–IV
5	$A(2, -1)$, $B(8, 7)$: the length, the midpoint and the perpendicular bisector	*6*	Q IV
6	Find where $y = 2x - 4$ meets $x^2 + y^2 = 16$, and the chord length	*6*	Q IV
7	A point 9 cm from a plane; obliques of 15 and 41 cm: the projections	*4*	Q III

HOW TO SPEND 45 MINUTES

Questions 1, 5 and 7 are quick — about *12* minutes for *14* marks. Do them first, then 3 and 6, and leave 2 and 4 last: a proof and a construction reward the time that is left, and punish the time that is not.

The year in one page

QUARTER	THE ONE IDEA	THE PICTURE
I — axioms and entry to space	three points determine a plane	the plane through three dots
II — parallelism	parallel to a line **in** the plane	a line and its parallel below
III — perpendicularity	perpendicular to two **intersecting** lines	the vertical mast
IV — coordinates	geometry becomes algebra	the circle and its chord



One box, and every angle of the year measured inside it.

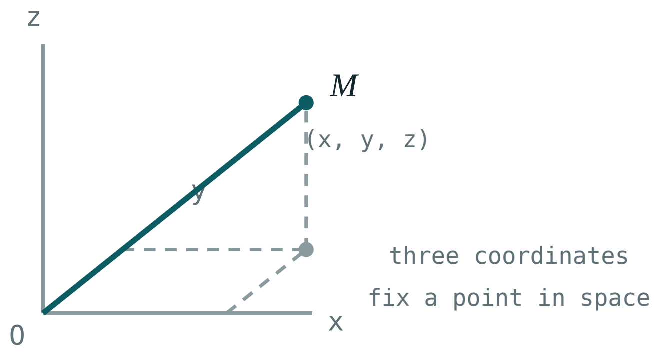
THE FOUR HABITS WORTH KEEPING

- draw the solid before naming anything;
- find the projection, and a right triangle appears;
- in coordinates, plot before you calculate;
- finish a proof with the sentence that states the conclusion.

The Grade 11 preview

Grade 11 geometry continues each of the four strands, and adds measurement.

GRADE 10	BECOMES, IN GRADE 11
points and lines in space	coordinates (x, y, z) and vectors in space
angles by projection	the scalar product, and angles without a picture
polyhedra and their sections	prisms and pyramids — surface area and volume
the circle in the plane	the cylinder, the cone and the sphere



The next step — a third coordinate, and every plane method carries over.

What to do over the summer. Nothing large. Keep the seven plane-geometry formulas and the four positions-in-space pictures somewhere you will see them, and in September the first lesson — the coordinate system in space — will read as an easy extension rather than a new subject.

ONE SENTENCE TO CARRY INTO GRADE 11

Every three-dimensional problem is solved by finding the one right triangle it contains. That was true in every lesson this year, and it stays true in every lesson of the next.

02 Worked examples

EXAMPLE 1 **Model answer, Q3: a cube of edge *8* — the angle between the space diagonal and the base.**

1. Projection of AC_1 is $AC = 8\sqrt{2}$.
2. $CC_1 = 8$
3. $\tan\theta = \dfrac{8}{8\sqrt{2}} = \dfrac{1}{\sqrt{2}}$
4. $\theta \approx 35.3°$

ANSWER

$\approx 35.3°$

EXAMPLE 2 **Model answer, Q6: where $y = 2x - 4$ meets $x^2 + y^2 = 16$.**

1. $x^2 + (2x - 4)^2 = 16$
2. $5x^2 - 16x = 0 \Rightarrow x(5x - 16) = 0$
3. $x = 0$ or $x = 3.2$
4. $(0, -4)$ and $(3.2, 2.4)$; chord $\sqrt{10.24 + 40.96} \approx 7.16$.

ANSWER

$(0, -4)$, $(3.2, 2.4)$; chord ≈ 7.16

EXAMPLE 3 **Model answer, Q7: a point *9* cm from a plane, obliques *15* and *41* cm.**

1. $\sqrt{225 - 81} = 12$
2. $\sqrt{1681 - 81} = 40$
3. Projections 12 cm and 40 cm.

   Both are Pythagorean triples.

ANSWER

12 cm and 40 cm

03 Interactive model

Give the class the paper’s mark allocation before they start; ten seconds of planning is worth five marks.

The year in twelve questions

A plane is determined by:

- two points
- three non-collinear points
- one line
- four points

Two lines that never meet are:

- parallel
- parallel or skew
- skew
- intersecting

A line ∥ to a plane is ∥ to:

- every line of it
- some line of it
- the normal
- nothing

Two planes are parallel if:

- one line of each is parallel
- two intersecting lines of each are parallel
- they do not meet at one point
- they look parallel

⊥ to a plane means ⊥ to:

- one line
- two intersecting lines
- two parallel lines
- the edge

The angle to a plane is measured to:

- the normal
- the projection
- any line
- the edge

A dihedral angle is measured:

- anywhere
- by rays ⊥ to the edge
- along the edge
- by projection

Each side of a section lies in:

- one face
- two faces
- the base
- none

S_proj equals:

- $S \cos \theta$
- $\frac{S}{\cos \theta}$
- $S \sin \theta$
- S

Perpendicular gradients satisfy:

- $m_1 = m_2$
- $m_1 m_2 = -1$
- $m_1 + m_2 = 0$
- $m_1 m_2 = 1$

Centre of $(x - 4)^2 + (y + 1)^2 = 25$:

- $(4, 1)$
- $(4, -1)$
- $(-4, 1)$
- $(-4, -1)$

A tangent line meets the radius at:

- $45°$
- $90°$
- $60°$
- any angle

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Annual control work	Yillik nazorat ishi	Годовая контрольная работа
Axiom	Aksioma	Аксиома
Parallelism	Parallellik	Параллельность
Perpendicularity	Perpendikulyarlik	Перпендикулярность
Section	Kesim	Сечение
Coordinate method	Koordinatalar usuli	Координатный метод
Solid of revolution	Aylanish jismi	Тело вращения
Volume	Hajm	Объём

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Which questions should be attempted first?

- the proof
- the quick ones — 1, 5 and 7
- the construction
- in order

Grade 11 begins geometry with:

- the sphere
- coordinates and vectors in space
- sections
- the circle

The one habit worth keeping is:

- memorising formulas
- finding the right triangle
- working fast
- skipping proofs

The scalar product will replace:

- the section method
- angles found by projection
- the axioms
- coordinates

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Three positions of two lines in space

   ANSWER Intersecting, parallel, skew

2. A plane is determined by

   ANSWER Three non-collinear points

3. Space diagonal of a cube of edge 8

   ANSWER $8\sqrt{3}$

4. Face diagonal of the same cube

   ANSWER $8\sqrt{2}$

5. Projection of a 15 oblique from a point 9 from the plane

   ANSWER 12

6. Midpoint of $(2, -1)$ and $(8, 7)$

   ANSWER $(5, 3)$

7. Length of that segment

   ANSWER 10

Medium · the standard question

7 PROBLEMS

1. Angle of the space diagonal of a cube with the base

   ANSWER $\approx 35.3°$

2. Dihedral angle of the plane ABC_1 with the base of a cube

   ANSWER $45°$

3. Perpendicular bisector of $(2, -1)$ and $(8, 7)$

   ANSWER $3x + 4y = 27$

4. Where $y = 2x - 4$ meets $x^2 + y^2 = 16$

   ANSWER $(0, -4), (3.2, 2.4)$

5. Chord length there

   ANSWER ≈ 7.16

6. Projection of a 41 oblique from a point 9 from a plane

   ANSWER 40

7. Diagonal section of a cube of edge 8

   ANSWER $64\sqrt{2}$

Hard · stretch and reason

7 PROBLEMS

1. Area of the section of a cube of edge 8 through A, B, C_1

   ANSWER $32\sqrt{2} \approx 45.3$

2. Angle between two space diagonals of a cube

   ANSWER $\approx 70.5°$

3. Distance from a vertex of a cube of edge 8 to the opposite space diagonal

   ANSWER $\frac{8\sqrt{6}}{3} \approx 6.53$

4. A tangent to $x^2 + y^2 = 16$ parallel to $y = 2x$

   ANSWER $y = 2x \pm 4\sqrt{5}$

5. The regular hexagonal section of a cube of edge 8: its area

   ANSWER $48\sqrt{3} \approx 83.1$

6. A rectangular box $9 \times 12 \times 20$: the angle of the space diagonal with the base

   ANSWER $\approx 53.1°$

7. Prove that the four space diagonals of a cube meet at one point and bisect each other

   ANSWER All pass through the centre

07 Homework

Homework — 4 tasks

This is the last geometry lesson of Grade 10. Keep the year-in-one-page sheet for September.

1. Rewrite in full every annual-paper question that lost a mark, naming the slip in the margin.
2. Write the year in one page: the four quarter ideas with one sketch each.
3. In a cube of edge 10 cm, find the angle between the space diagonal and the base, and the area of the diagonal section.

4. Write one sentence saying what you will keep from this year's geometry, and one saying what you will practise before September.